



CSE-476: Introduction to Natural Language Processing
Quiz 1

Multiple Choice Questions (Single correct)

Instructions: Select the correct option with a short explanation.

- Q1. In BIO tagging for NER, which tag marks tokens that are not part of any entity?
- (A) Beginning of an entity
 - (B) Inside an entity
 - (C) Outside any entity
 - (D) Between two tokens
- Q2. In sequence tagging, which statement best describes a model that ignores labels and uses only the observed words?
- (A) It uses only words (observations) and ignores labels
 - (B) It is identical to an HMM
 - (C) It models transitions between labels but does not consider the words
 - (D) It models emissions $P(\text{word} \mid \text{label})$
- Q3. In a Hidden Markov Model (HMM) for NER, a transition probability is:
- (A) $P(\text{label} \mid \text{previous label})$
 - (B) $P(\text{label} \mid \text{word})$
 - (C) $P(\text{word} \mid \text{previous word})$
 - (D) $P(\text{word} \mid \text{label})$
- Q4. Which of the following best describes a system that assigns grammatical roles like subject and object based purely on syntax?
- (A) A task that identifies predicates in a sentence and labels roles of elements such as agent, theme, or location
 - (B) A process that assigns grammatical roles like subject and object based on syntactic structure
 - (C) A method that classifies named entities and links them to a knowledge base
 - (D) A technique that labels words with their parts of speech and morphological features

- Q5. Which of the following best describes a simple tokenizer that splits text only on spaces and punctuation marks?
- (A) A technique that compresses text by replacing consecutive repeated characters with a count and a symbol
 - (B) A tokenization method that iteratively merges the most frequent pair of symbols in a corpus
 - (C) A tokenization method that splits text based on whitespace and punctuation
 - (D) A character-level encoding scheme that assigns fixed-length binary codes to characters
- Q6. Which of the following best describes the main goal of part-of-speech tagging in NLP?
- (A) To label each token with a grammatical category
 - (B) To convert text into a sequence of symbols suitable for modeling
 - (C) To reduce vocabulary size to zero
 - (D) To assign probabilities to sentences
- Q7. Why is splitting text by spaces (`text.split(" ")`) problematic if it treats every punctuation mark as its own token?
- (A) It produces too many tokens
 - (B) It does not work well for languages without explicit word boundaries
 - (C) It cannot tokenize English at all
 - (D) It ignores punctuation entirely
- Q8. Which of the following is typically treated as a document-level classification problem rather than a sequence tagging task?
- (A) Part-of-speech tagging
 - (B) Named Entity Recognition
 - (C) Dependency parsing
 - (D) Sentiment analysis
- Q9. Which of the following is a misconception about BIO tagging that is actually not a real limitation?
- (A) It cannot represent overlapping or nested entities
 - (B) It requires too many labels
 - (C) It does not support multiple entity types
 - (D) It ignores entity boundaries

- Q10. When choosing between a very fast but less accurate model and a slower but more accurate one, which trade-
- (A) Speed vs accuracy
 - (B) Tokenized sequence length vs vocabulary size
 - (C) Precision vs recall
 - (D) Memory vs computation

End of Paper